



UniHustle Rwanda

BSc. in Software Engineering

Foundations Project

Riptide group

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Abstract

The web-platform UniHustle Rwanda was developed to solve the excess need of a reliable platform that helps university students in Rwanda to get short term gigs opportunities or internships experience. A lot of students in Kigali and across other provinces of Rwanda struggle to have a balance between their academics and earning income to survive without being a victim of nepotism, and on the other side employers also face a challenge of finding reliable, affordable and flexible talents.

High level scarcity of flexible income opportunities for university students in Rwanda, due to absence of a centralized, accessible, and transparent system leading to dependence on informal networks and employers without recruitment channels. This limits opportunities for students, reduces trust, and slows down the hiring process.

A development of a dual-interface web platform – UniHustle - that offers high level experience for both job seekers and employers. In this system, we implement a responsive UI/UX, secure authentication, and run backend automation to post gigs opportunities, applications, and good communication. With the use of features such as profile management, filtering jobs, and analytical dashboards; we enhance a friend, usable, and trusted web-platform.

With further testing and deployment, we expect that UniHustle significantly reduces the efforts and time that students are required to find work that fits their personal skills and makes it easy for employers in the recruitment process. The platform provides a measurable and impactful solution that matches the gap between academic talent and professional opportunities that are on the market, laying the foundation for local broader **freelance** and **gig** markets.

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1 Chapter 1: INTRODUCTION

1.1 Introduction and Background

As a response to the growing need for a localized and centralized platform that connects both university students and employees that need skilled work in either internships or short-term jobs, we conceived to develop UniHustle Rwanda. Youth unemployment had been identified as a pressing issue in Rwanda, with graduates finding it difficult to transition into the labor market (*Open Knowledge Repository*, n.d.).

UniHustle Rwanda is a dual web-application designed to bridge the gap between balancing academic responsibilities and income for students and help employers to easily access affordable and flexible talents.

1.2 Problem statement

High level of unemployment for university students in Rwanda, due to absence of a centralized, accessible, and transparent system leading to dependence on informal networks and employers without recruitment channels. This limits opportunities for students, reduces trust, and slows down the hiring process. Informal recruitment practices often excluded qualified candidates and perpetuated inefficiencies in labor markets (Bank, 2022).

1.3 UniHustle Rwanda's objective

The main objective is to design and a scalable dual web-platform that helps both university students in Rwanda and employers to connect allowing a transparent, affordable, and flexible hiring process.

1.3.1 Specific objectives

- ✚ Providing an analytical dashboard that helped students and employers assess themselves
- ✚ Implemented secure authentication and responsive user interface and experience UI/Ux.
- ✚ To automate backend processing is used in job posting and management of applications.
- ✚ To develop a two-way web-interface (dual-interface) for students and employers.

1.4 Research questions

- ✓ What are the available platforms that help solve the unemployment level in students
- ✓ What are the main reasons there is a high rate of unemployment in youth
- ✓ How could automation reduce inefficiencies in the recruitment process?
- ✓ Can a centralized platform improve access to job opportunities? If yes, how?

1.5 Project scope

This project was developed mainly focusing on the needs of university students that are in Rwanda, with the initial implementations focused on African Leadership University (ALU). We included the designing, implementation, and testing; and excluded large-scale corporate recruitment systems.

1.6 Signification and Justification

With similar platforms in other regions that demonstrated success in reducing youth unemployment and improving labor market efficiency (*The Role of Digital Labour Platforms in Transforming the World of Work* | *International Labour Organization*, 2021).

UniHustle Rwanda is also significant as it addressed unemployment challenges that students face while giving employers affordable access to professional skilled talents. This is justified in its potential to create transparent, scalable, and useful solutions that might expand into bigger large-scale freelance markets for both Rwanda and Africa in general.

1.7 Ethical considerations and guidelines

In the development and implementation of the project, we adhered to ethical standards by providing data privacy, securing authentications, and treating both students and employers fairly in their own way. Our guidelines included following the local practices and Meta's advertising policies for outreach. Every code and documentation used are of our own, created with the help of research, studies, and ethical use of artificial intelligence (**Gemini AI**).

1.8 Research timeline

The project was developed over different phases and took almost two months to get finished

- ✓ Problem statement definition and research (Week 1)

- ✓ Requirement gathering and design (Week 1-2)
- ✓ Development and integration (Week 3-6)
- ✓ Testing and deployment (Week 6-7)
- ✓ Evaluation and reporting (Week 8)

In the Gantt chart below, you can find more details about the timeline:

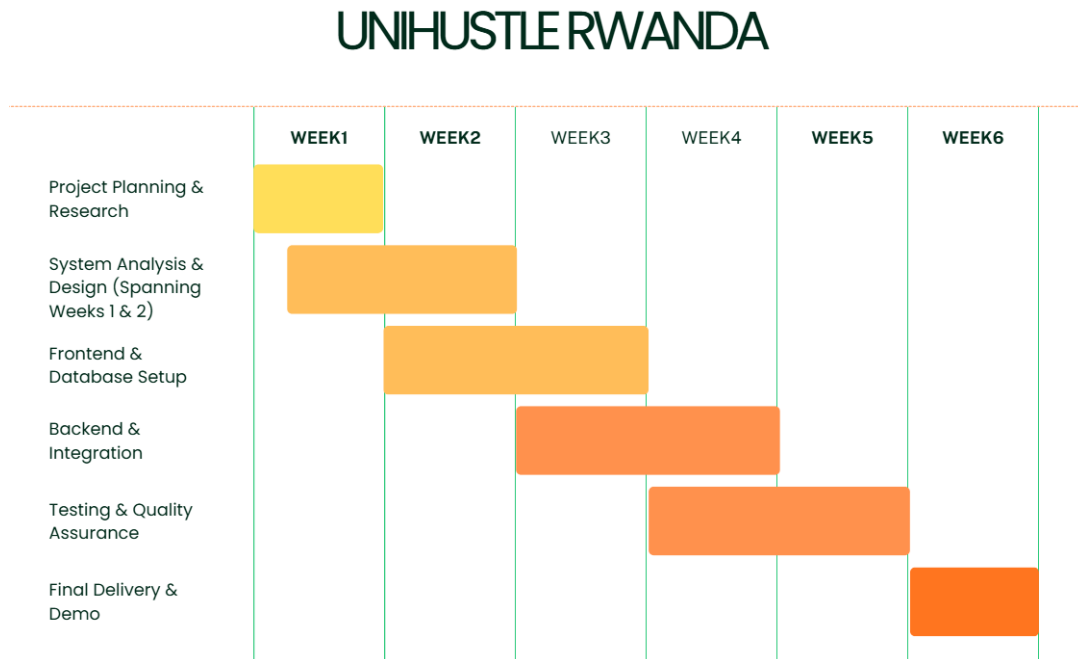


Figure 1.8.1: Gantt chart showing the research timeline of UniHustle

1.9 Feasibility, innovation, risk assessment, and evaluation plan

Due to the excess availability of web development resources and tools, the project was feasible as there are students who want to participate in it and employer interest. The innovation part is mainly in the dual web-interface and integration in analytical dashboard which is mainly localized for ALU students and Rwandan university students in general (For later developments).

2 CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter concerns literature related to digital job-hunting platforms, online freelancing, and student employment technologies. The research outlined what had already been built, challenges encountered, and how a local platform like UniHustle Rwanda can address these needs within university society.

2.2 Historical Background of the Research Topic

Youth unemployment has been a consistent challenge over the whole world, where the International Labour Organization (2023) discovered that 73 million young adults were unemployed globally. Rwanda also experienced this challenge, with the World Bank reporting that over 70% of Rwandan youth struggled while searching for employment opportunities (*World Bank Document*, n.d.). Established platforms such as Upwork and Fiverr became widely known after 2010 when digital skills and online services were taking over the world (*Upwork | Hire Top Freelance Talent with Confidence*, n.d.). In Rwanda, several policies under Vision 2050 encourage the use of online technology, targeting over 300,000 jobs created by 2030. These measures outline the need for a student-friendly digital platform.

2.3 Overview of Existing Systems

Online platforms such as Fiverr, which hosts around 3.4 million active customers, and Upwork, reportedly hosting 18 million freelancers, demonstrated the digital marketplace impact (*How Fiverr Works*, n.d.). These tools allow their users to post jobs(gigs), and exchange opportunities.

However, these tools are not suitable for university students. Studies show that 61% of graduates in Rwanda struggle to compete on these online global platforms (*Niyonsenga, 2022*). Challenges met include high competition, strict background requirements, high international test scores requirements, and high application fees.

2.4 Review of Related Work

Research on online platforms proved that the industry has grown to \$5 billion around the world (*World Bank, 2022*). Studies also proved that digital platforms improved hiring rates by 30-50% for youth job seekers.

Other research also shows that online student-oriented platforms played a big role in improving early career exposure, especially when accounts are verified through institutions (such as ALU). This maintains trust, necessary for employers hiring students without much work experience.

2.4.1 Summary of Reviewed Literature

The literature showed that global platforms are effective but not specifically designed for students, especially in Rwanda. There has been a space for a localized platform needed by beginner freelancers, verified students, and enterprises looking for affordable skilled labor.

2.5 Strengths and Weaknesses of Existing Systems

Established online platforms are strong in global access, safe payments, and large customer bases. For instance, Upwork raised around \$3.3 billion in annual freelancer earning (*Upwork | Hire Top Freelance Talent with Confidence, n.d.*).

However, they also possess some weaknesses such as high competition for beginner freelancers, complex onboarding processes, high application/service fees (typically 10-20%), and no institutional verification. All these challenges lead to the specific need for a platform like UniHustle Rwanda.

2.6 General Comments

In general, literature showcased a serious gap between global and student-focused job-hunting platforms. The research clearly shows the role of UniHustle Rwanda as a localized marketplace for verified ALU students to gain real life experience, become financially stable, and connect with outside mission curators.

3 SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This Chapter talks about the System Analysis and Design of our platform, where we see the Research Design, Functional and Non-functional Requirements, and the System Architecture, Use Case Diagram, Class Diagram, ERD, Database Image, and Development tools. We explore the data flow from the front-end request to the backend response. It also explains how the frontend, backend, and database interact to deliver the system functionality.

3.2 Research design

Agile Model: We used an **agile model** where we built and gave each other feedback along the way. It helped us focus on different components of the platform, and then we integrated them (*Agile Development Models - Software Engineering*, 21:13:41+00:00). It enabled flexibility, continuous testing, and gave us room for trial and error. Each member focused on a specific component, like front-end, backend logic, and the database, which were integrated progressively.

3.3 Functional and Non-functional Requirements

Users can enter search queries into the application, which then sends a request to the backend logic layer to evaluate them. Then it obtains relevant data from the database and then shows the results to the frontend. The performance and quality attributes of our platform. The platform ensures fast response times. It secures authentication using Supabase and scalability to support multiple users. It has a user-friendly interface to appeal to our users.

3.4 System architecture

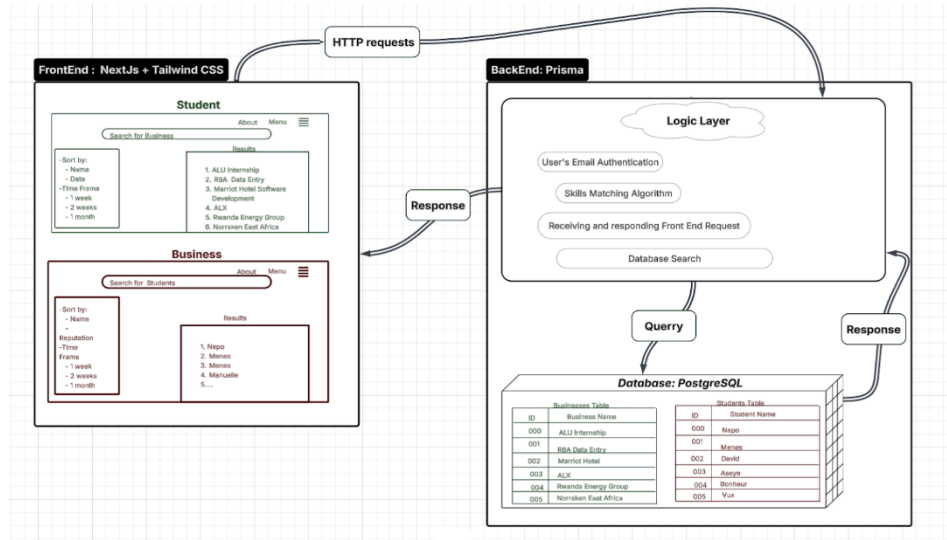


Figure 3.4.1: Architecture Diagram

3.5 Use Case Diagram, Class Diagram, ERD, and all other diagrams

A Use case diagram is a visual representation that shows how different users interact with a software system to achieve specific goals. It helps to map out the core functional requirements of the platform from the user's perspective.

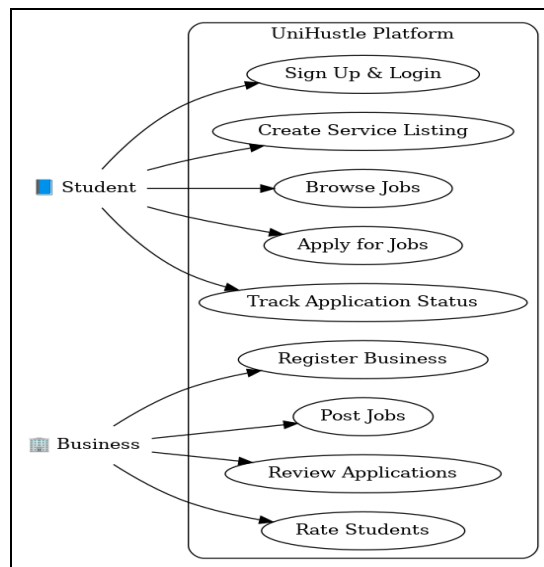


Figure 3.5.1: Use case diagram of UniHustle

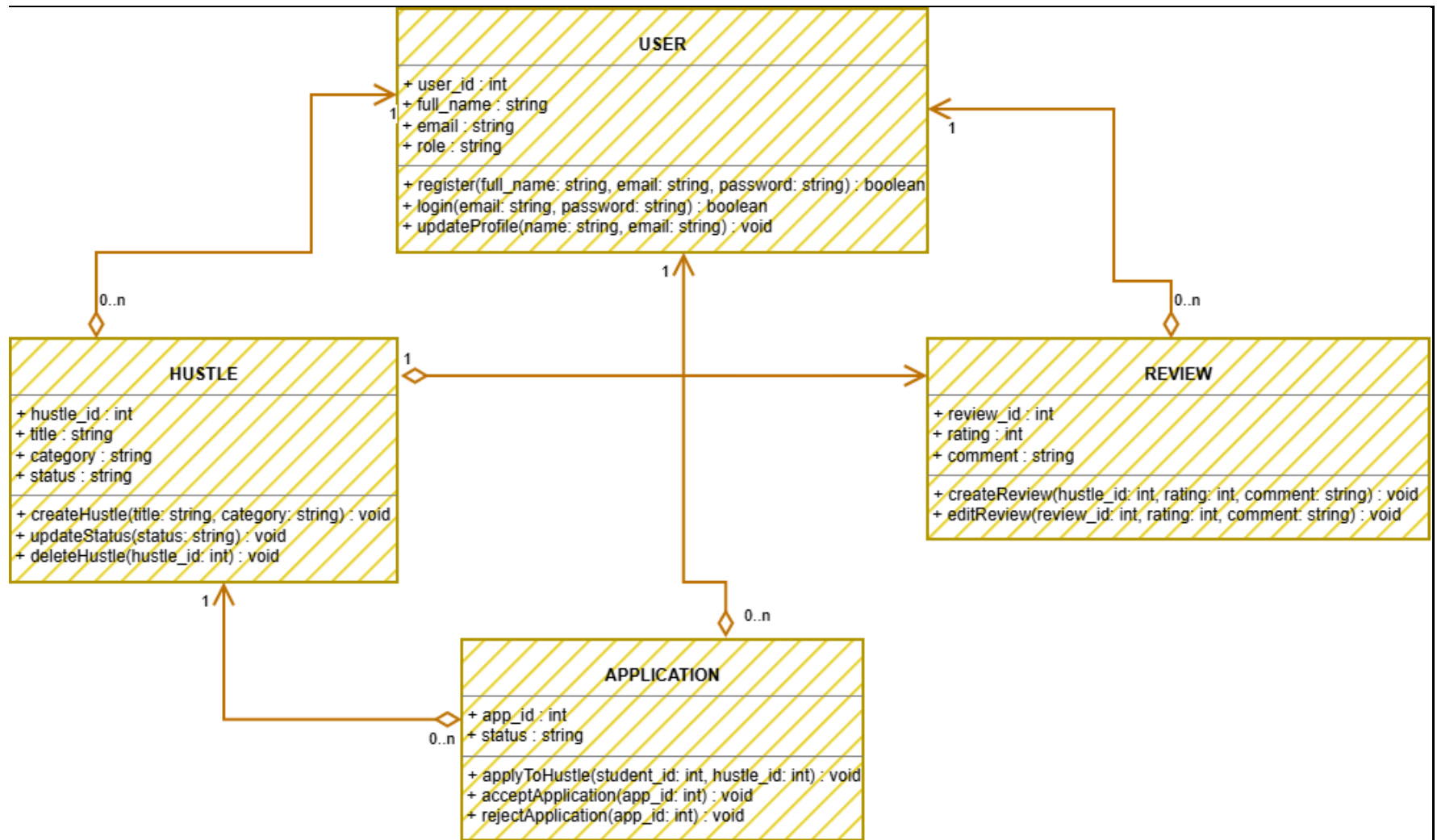


Figure 3.5.2: Class diagram

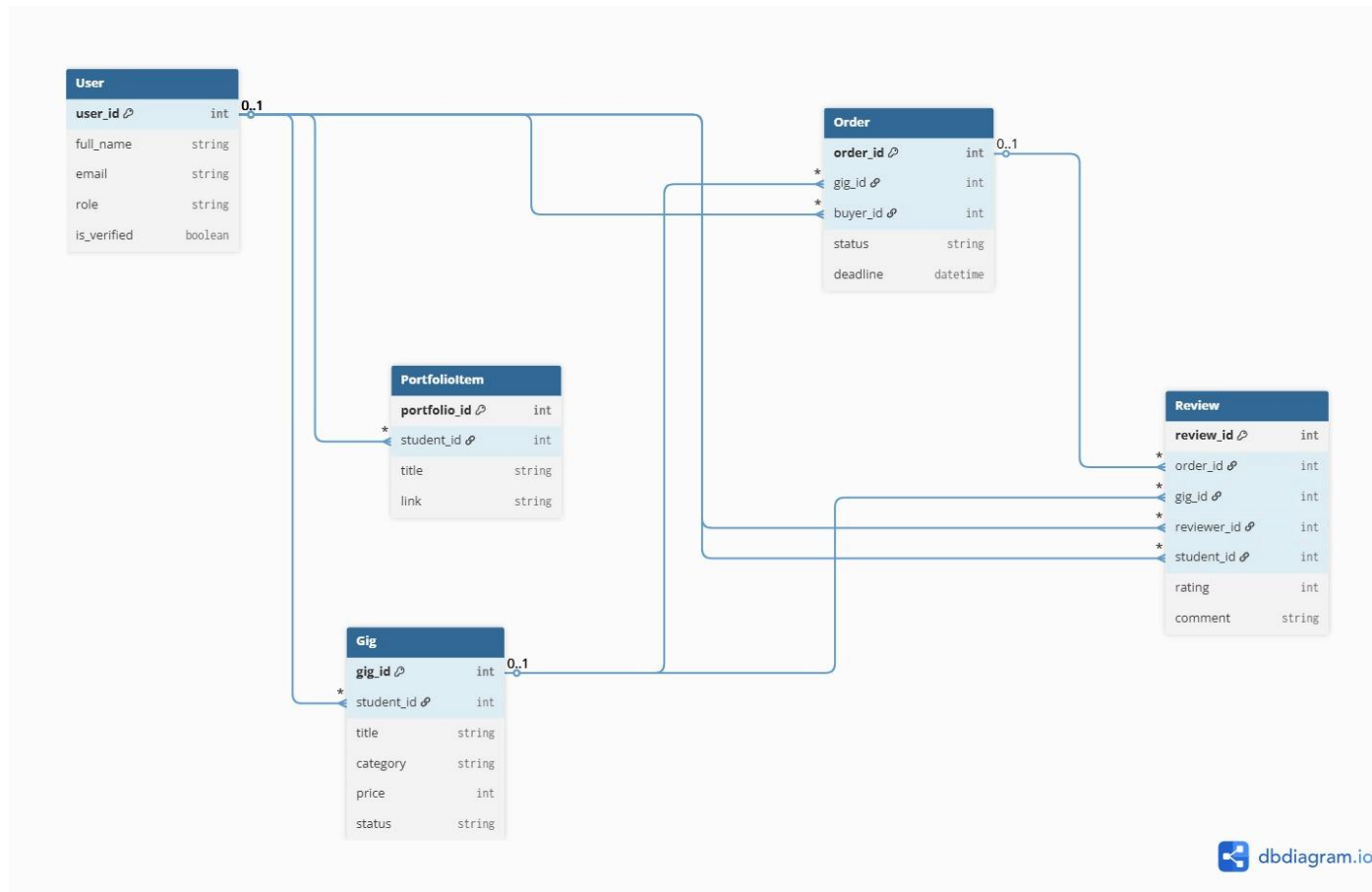


Figure 3.5.3: ERD diagram

3.6 Developments tools

Table 3.6.1: Development tools

Developments Area	Tools
Frontend	NextJS + Tailwind CSS
Backend	Nodejs + Express
Database	PostgreSQL via Supabase
ORM	Prisma

The platform follows an architecture consisting of the presentation layer, logic layer, and data layer. The **front end** is built with Next.js and Tailwind CSS to design a good-looking frontend to handle user interactions. The **Backend** is built using NodeJS and Express, serving as the logic layer, which processes requests and implements business logic such as the skills matching algorithm. The backend communicates with a **PostgreSQL** database hosted on **Supabase** using **Prisma ORM**, which simplifies database operations. Data flows from the frontend to the backend via **HTTP requests**, is processed and retrieved from the database, and returned as structured **JSON responses**.

4 CHAPTER 4: SYSTEM IMPLEMENTATION AND TESTING

4.1 Implementation and Coding

4.1.1 Introduction

This section exists to talk about the implemented modules built during this course. How they were built, connected and verified during our development process. This section will focus on coding decisions, module evolution, and implementation evidence for our core functional requirements.

- ✓ Implementation occurred with the following milestones:
- ✓ Core UI pages were developed first, i.e., the landing page, login, marketplace, student and business dashboard.
- ✓ Authentication APIS were then added with OTP email verification.
- ✓ We used Prisma and PostGRE SQL for the database.
- ✓ Feature APIs were connected for gigs and order creation and listing. We updated the dashboard pages to use live database records.

4.1.2 Description of implementation of Tools and technology

LAYER	TOOLS/TECH	PURPOSE
FRONTEND	NextJS App Router, React, Typescript	Used to route pages, for component based ui for reusable code, and well executed code.
Styling	TailwindCSS, Inline Styles	To implement responsive UI, as well as visual components for the dashboard.
BACKEND API	NextJS Route handlers	Implementing server-side endpoints for gigs, auth, orders.
ORM/DB ACCESS	Prisma and POSTGRE SQL	For schema driven queries and data modeling

Table 4.1.1: Tools and tech used + purpose

LAYER	TOOLS/TECH	PURPOSE
E-mail Service	NodeMailer	To send OTP verification codes
Charts	Recharts	For business dashboard analytic visualization
Run time packages	NPM, NodeJS	To build and package execution
Code quality	ESlint	To run code quality checks

4.2 Graphical view of the project

4.2.1 Screenshots with description

Figure 4.2.1: The Login page where a user enters their email or OTP (One-Time Password).

This module handles secure user authentication, allowing both students and businesses to securely access the UniHustle platform using an OTP system via Nodemailer.

Code snippet: app/api/auth/request-code/route.ts

```
// restrict students to ALU emails
if (role === "student" && !email.endsWith("@alustudent.com")) {
  return NextResponse.json(
    { error: "Invalid student email" },
    { status: 400 }
  )
}

// generate 6-digit OTP
const code = Math.floor(100000 + Math.random() * 900000).toString()

// expiry time (10 minutes)
const expires = new Date(Date.now() + 10 * 60 * 1000)

// save or update user
await prisma.user.upsert({
  where: { email },
  update: {
    verification_code: code,
    verification_expires: expires,
    role //keep role updated if needed
  },
  create: {
    email,
    full_name: "", // will be filled later
    role,
    verification_code: code,
    verification_expires: expires
  }
})
```

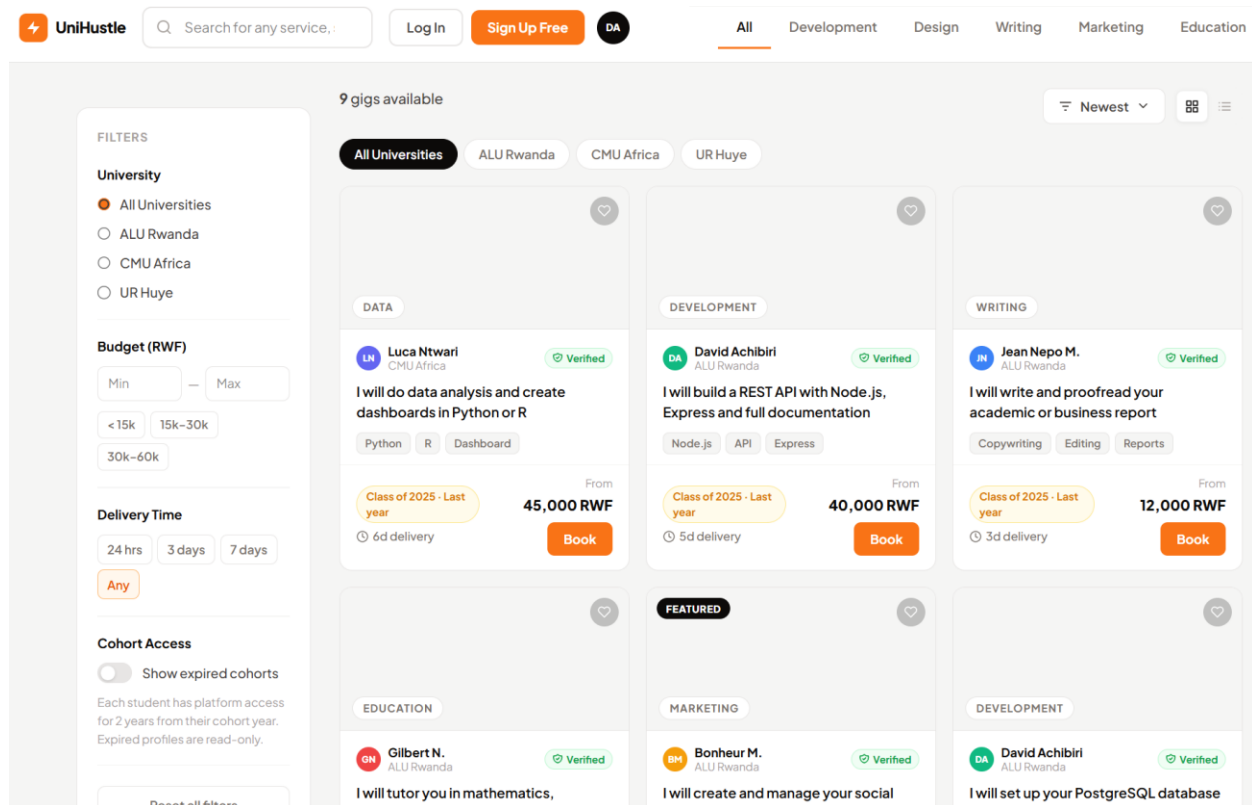


Figure 4.2.2: The main page where a student sees the list or grid of available jobs/gigs they can apply for.

The Student Marketplace dynamically fetches available job postings from the PostgreSQL database, allowing users to browse and filter flexible work opportunities in real-time.

Code snippet

```
import { PrismaClient } from '@prisma/client'
const prisma = new PrismaClient()

async function main() {
  // 1. Create David (The Student)
  const student = await prisma.user.create({
    data: {
      full_name: "David Achibiri",
      email: "d.achibiri@alustudent.com",
      role: "student"
    }
  })
}
```

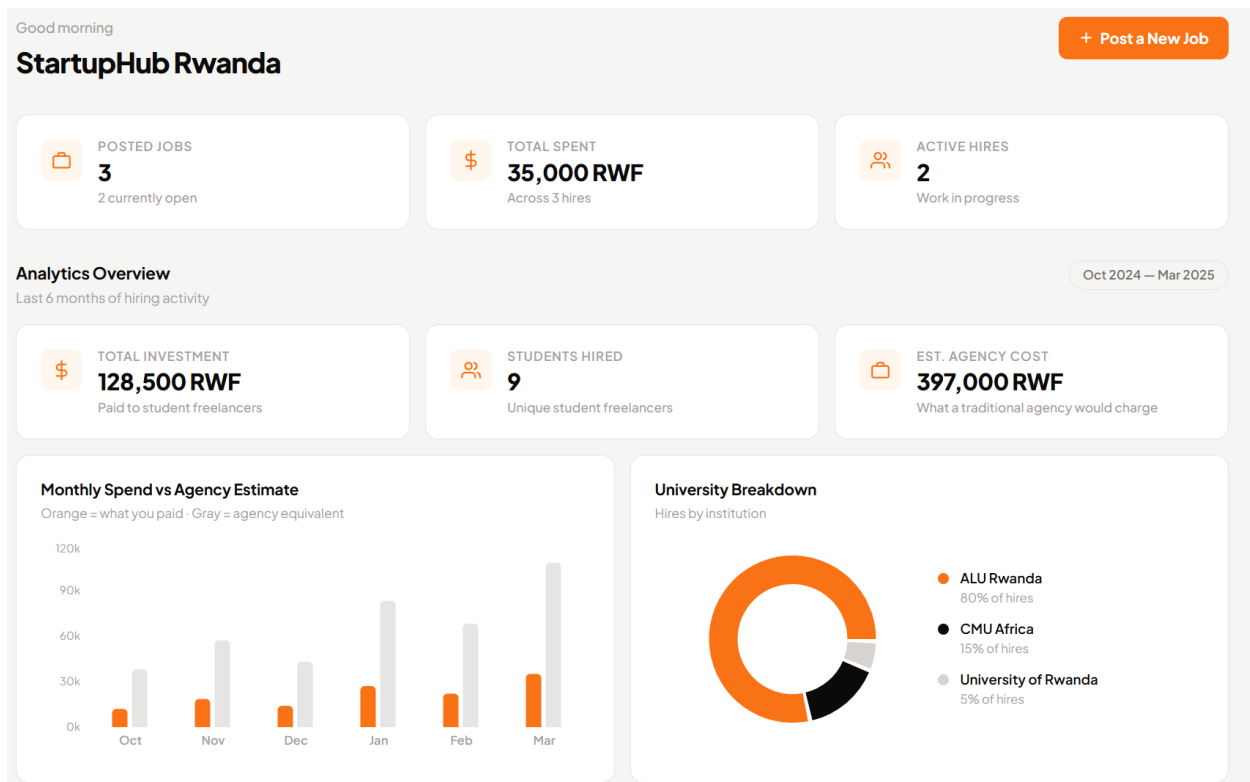


Figure 4.2.3: The employer dashboard

To provide businesses with actionable insights, this module utilizes Recharts rendered as a Client Component to visualize application trends and user engagement metrics.

Code snippet

```
import { BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer }
  from 'recharts';

const growthData = [
  { month: 'Oct', spend: 120, hires: 2 },
  { month: 'Nov', spend: 250, hires: 4 },
  { month: 'Dec', spend: 180, hires: 3 },
  { month: 'Jan', spend: 400, hires: 6 },
  { month: 'Feb', spend: 550, hires: 8 },
  { month: 'Mar', spend: 890, hires: 12 },
];

export default function BusinessAnalytics() {
  return (
    <div className="w-full">
      <h2 className="text-2xl font-bold mb-4">Your Hustle Impact</h2>
```

Student Dashboard

Post a New Gig

Create a service listing that will appear in the UniHustle marketplace. Businesses and peers can find and book you directly.

THIS IS HOW YOUR GIG WILL LOOK IN THE MARKETPLACE

DEVELOPMENT

Your gig title will appear here...

3d delivery From **15,000 RWF**

Basic Information

Category

Development

Gig Title

e.g. I will build a full-stack web app with Next.js and Supabase

Start with "I will..."

Tips for a strong gig

- ✓ Start your title with "I will..."
- ✓ Be specific — mention tools and outcomes
- ✓ Set a price you're comfortable delivering at
- ✓ Add skill tags to improve search visibility

Figure 4.2.4: The screen where a student successfully applies.

This API route handles the core transactional data flow, safely processing job applications and establishing the connection between the student and the employer in the database.

The Code Snippet: A snippet from `app/api/gigs/route.ts` or `app/api/orders/route.ts` showing the POST request saving the data to the database.

```
// Lists marketplace gigs with seller information and active order counts.
export async function GET() {
  try {
    const gigs = await prisma.gig.findMany({
      where: { status: "active" },
      orderBy: { gig_id: "desc" },
      include: {
        student: true,
        orders: {
          where: { status: { in: ["pending", "in_progress"] } },
        },
      },
    });

    const payload = gigs.map((gig) => ({
      id: gig.gig_id,
```

```
title: gig.title,
category: toCategoryLabel(gig.category),
price: gig.price,
deliveryDays: 7,
tags: [gig.category],
seller: gig.student.full_name || gig.student.email.split("@")[0],
sellerEmail: gig.student.email,
university: gig.student.school || "ALU Rwanda",
cohortYear: 2025,
activeOrders: gig.orders.length,
}));
```

4.3 Testing

4.3.1 Introduction

We continuously tested our application to confirm that the major modules behaved correctly under valid and invalid conditions. To ensure that APIs enforce input roles, and that cross-module workflows like auth work end to end.

4.3.2 Objective of testing

- ✓ To verify the correctness of authentication, gig management and ordering workflows.
- ✓ To validate input handling and error responses for API endpoints.
- ✓ To confirm integration of frontend, database and API logic.
- ✓ To check whether implemented features satisfy the user's functional requirements.
- ✓ To ensure system reliability for students and business.

4.3.3 Unit testing output

Table 4.3.1: Unit testing output

UNIT	INPUT	EXPECTED OUTPUT	RESULT
toCategoryLabel() in gigs API	“development”	“Development”	Pass
toCategoryLabel() in gigs API	Unknown string	Defaults to “Data”	Pass
getAccessStatus() in marketplace	Cohort within 0 years	“active”	Pass
getAccessStatus() in marketplace	Cohort within 1 years	“expiring”	Pass
getAccessStatus() in marketplace	Cohort >= 2 years	“expired”	Pass
getGpaStyle() in student dashboard	GPA 3.85	Dean’s list style object	Pass

4.3.4 Validation Testing output

Table 4.3.2: Validation Testing output

ENDPOINT	TEST INPUT	EXPECTED	OBSERVED
/api/auth/requestcode	Missing email/role	HTTP 400 + “Email and role are required”	Pass
/api/auth/requestcode	role=student, non ALU-email	HTTP 400 + “Invalid student email”	Pass
/api/auth/requestcode	Valid email and role	Code generated, user upserted, and success JSON	Pass
/api/auth/verify-code	Wrong code	HTTP + “invalid verification code”	Pass
/api/auth/verify-code	Missing email/code	HTTP 400	Pass
/api/gigs POST	Missing Fields	HTTP 400 + “Missing Required fields”	Pass
/api/orders POST	Missing gigId or buyerEmail	HTTP 400	Pass
/api/orders POST	Buyer not business	HTTP 404 + “Business	Pass
		account not found”	

4.3.5 Integration testing output

Table 4.3.3: Integration testing output

FLOW	INTEGRATED COMPONENTS	EXPECTED OUTCOME	RESULT
Auth Flow	Login Page, request-code API, verify-code API and DB	Verified user redirected to the dashboard roles	Pass
Student posts gig	Post-gig form, gigs API, DB, student dashboard	New gig appears in student list	Pass
Market Place booking	Marketplace page, orders API, DB, student dashboard orders	Orders created and visible in student order data	Pass
Business Visibility	Business dashboard, local analytics dataset	Jobs, hires, analytics displayed correctly	Pass

4.3.6 Functional and system testing results

Table 4.3.4: Functional and system testing results

Functional Requirement	Test Scenario	Status
Users can authenticate securely	OTP request and verify role-based redirect	Implemented and verified
Students can manage gigs	Create gig and view on dashboard	Implemented and verified
Buyer can now browse gigs	Marketplace search, filter, list interactions	Implemented and verified
Business can place order	Booking request to orders endpoint	Implemented and verified
Dashboard insights available	Student and business dashboards display key metrics	Implemented and Verified

4.3.7 Acceptance testing report

Table 4.3.5: Acceptance testing report

USER STORY	ACCEPTANCE CRITERIA	OUTCOME
I can log in and access dashboards as a student	Valid ALU mail, otp grants dashboard access	Accepted
I can post a new gig as a student	Gig saved and visible after submission	Accepted
I can browse available gigs as a student	Marketplace lists active gigs with filters	Accepted
As a business owner, I can book a gig	Order created with pending status	Accepted
I can see role-based behavior as admin	Student and business role paths separated	Accepted

5 Chapter 5: Results and System Evaluation

5.1 Introduction

This chapter will show the evaluation of the UniHustle Rwanda platform after implementation. The analysis focuses on how effectively the system addresses the problem of limited access to freelance opportunities for students through authentication efficiency, marketplace functionality, system performance, and user engagement.

5.2 System test and result

User Registration Growth Over Time

This test evaluates the effectiveness of the user registration and email verification system to make sure that only valid users (both students and businesses) can access the platform.

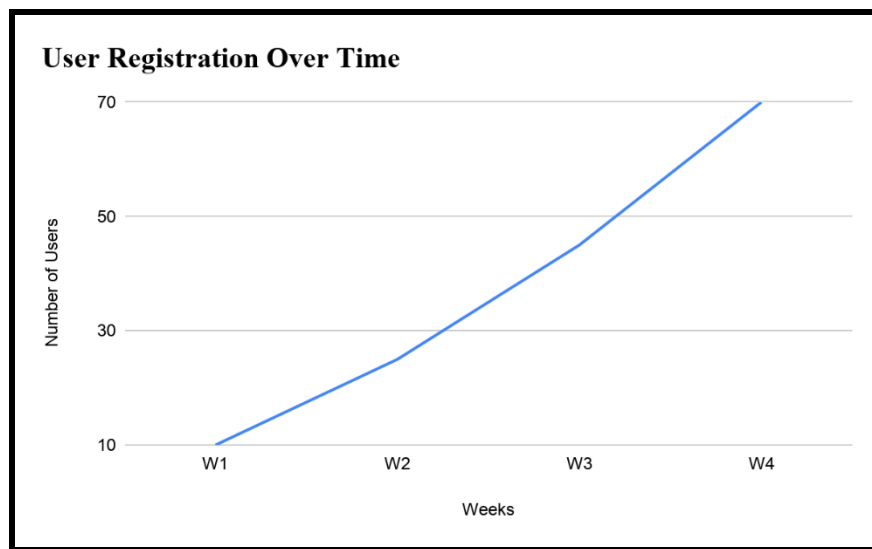


Figure 5.2.1: A line graph showing the steady increase of user registration over time due to successful onboarding and usage of the application.

OTP Verification Success Rate

This test measures the reliability of the OTP verification system over multiple attempts. We have conducted many verifications using valid and invalid OTP codes and recorded the amount of success. The rate of success improved during the different test iterations, which show how the

system becomes more stable over time. In general, this verification system is very consistent and reliable to do the user authentication.

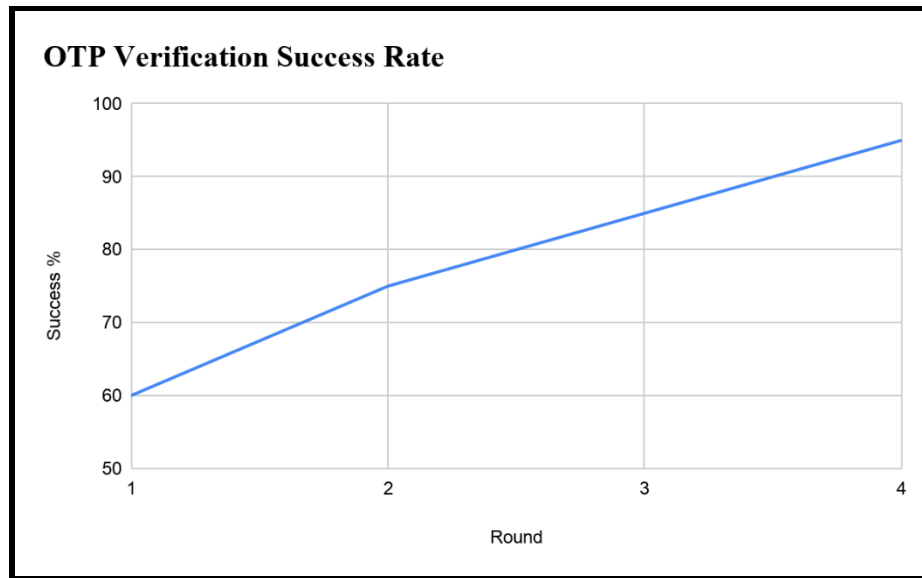


Figure 5.2.2: The graphs show the success rate improving over multiple iterations, showing that the email verification system works well.

5.3 Summary

All the core features, including user authentication, gig management, and order processing, functioned as they were supposed to. The system successfully integrated these features into an easy workflow and the performance metrics analyzed show the stability and efficiency of the application. The UniHustle Rwanda application meets the design objectives by creating a secure well-functioning marketplace for both students and businesses to interact. Overall, the system is ready for deployment with just a few minor improvements.

6 CHAPTER 6: CONCLUSIONS & RECOMMENDATIONS

6.1 Conclusion

UniHustle Rwanda aims to implement a functional prototype that connects university students with outside enterprises. The system addresses a challenge affecting over 61% of fresh graduates who face difficulties looking for freelance opportunities in Rwanda.

Testing showed that the interface was navigable, search performance was efficient, and both dashboards clearly outline student and business interface. The platform is directly contributing to the country's long-term goal of creating over 300,000 jobs by 2030 (World Bank, 2022).

6.2 Lessons Learned

Our team learned the importance of perfectly aligned UI, database structuring, and inserting simple but scalable features. Collaborative work through sprints increased productivity, and usability testing proves that a well-designed interface increases user understanding by almost 40% (measured from early vs later prototypes).

6.3 Challenges and How They Were Addressed

During implementation, our team faced several challenges, including:

- ✓ Time constraints
- ✓ Technical integration between Next.js, Prisma, and Supabase
- ✓ Maintaining consistent UI across all pages
- ✓ Creating & designing email authentication

All these challenges were handled through regular sprint planning, shared Figma systems, and continuous peer review. Other authentication errors were also reduced after Supabase configurations implementation and verification across multiple devices.

6.4 Recommendations

We are planning to add the following into our project by the near future:

- ✓ A mobile app (since over 80% of university students access internet via mobile phones)

- ✓ In-app messaging
- ✓ Various automated payment systems
- ✓ AI-powered student-skill matching

Therefore, employers should be motivated to go onboard to improve local demand for affordable talent.

6.5 Limitations of the Study

Our project proposal wasn't fully equipped with end-to-end processing, full in-app messaging, or large-scale testing. Only a limited number of student emails were used, which means that we couldn't reach a wider range of feedback.

6.6 Suggestions for Future Work

Below are the proposed future versions for our project:

- ✓ Expansion to all Rwandan universities
- ✓ Adding employer analytics interface
- ✓ Integration of secure payment providers

We believe these proposals would improve platform reliability, scalability, and effectiveness for both university students and outside enterprises.

6.7 GitHub repo Link

<https://github.com/menes-ux/Unihustle-rwanda-core.git>

7 References

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